

Sustainability Alignment and Structure in the Global Nickel Trade Network: A 2024 Network Analysis

GROUP 3

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I. ABSTRACT

This study investigates the structure of the 2024 global nickel trade network, a critical backbone of the green energy transition. Despite nickel’s central role in electric-vehicle (EV) batteries, its production often entails substantial environmental costs. Using a dual-method network approach, this paper examines whether sustainability performance shapes trade partnerships and whether leadership in clean-technology production translates into greater nickel demand.

First, Multiple Regression Quadratic Assignment Procedure (MRQAP) results show no evidence that environmental similarity—measured by the Environmental Performance Index—significantly predicts bilateral nickel trade once standard gravity controls are included. Instead, geographic distance and economic size remain the dominant drivers of trade flows.

Second, Exponential Random Graph Models (ERGM) reveal that the network is structured by export centralization, with a small core of dominant suppliers benefiting from preferential attachment. Moreover, a clear functional separation emerges in industrial demand: while battery export capacity significantly predicts nickel imports, EV export capacity does not. This suggests a decoupling between raw material sourcing and final vehicle assembly within the green-technology supply chain.

Overall, the findings indicate that the green transition’s upstream supply chains continue to operate according to conventional economic logic rather than sustainability alignment, highlighting risks of greenwashing and underscoring the need for greater supply chain transparency.

II. INTRODUCTION

The global transition toward electrification and renewable energy has driven an unprecedented demand for critical minerals, with nickel playing a central role in the composition of high-capacity batteries, used commonly in electric vehicles (EVs), where most market growth is occurring (IEA, 2024a). The global nickel market is expanding rapidly to meet this demand, but its supply chains remain unstable, affected by sudden shocks and significant shifts in production and processing capacity (IEA, 2024b). This establishes the global nickel trade as a critical connection for the green transition, linking the production of low-carbon technologies to the mining of raw materials.

This linkage, however, presents a significant contradiction. While nickel is essential for "green" technologies, its extraction and refining are well known for being energy-intensive and polluting processes (Wei et al., 2020; Institutional Shareholder Services, 2023). The problem is concentrated in the *type* of processing used in major producer nations. A large portion of new supply, particularly from Indonesia, comes from laterite ores processed into low-grade Nickel Pig Iron (NPI). This method is exceptionally carbon-intensive; research quantifies that producing one ton of NPI emits a massive 69 tCO₂-eq, compared to just 14-18 tCO₂-eq for other, higher-grade nickel products (Wei et al., 2020). This high emission rate is fueled by a heavy reliance on "captive" coal-fired power plants built specifically for these refineries (IEA, 2024b).

This creates a basic conflict: the rising demand for a sustainable energy future may be simultaneously driving unsustainable environmental degradation in producer nations. This degradation is not just a global carbon problem; it is a severe local one. The mining and refining processes are linked to widespread environmental damage. For instance, water pollution has been identified as the most frequent controversy associated with nickel mining (Institutional Shareholder Services, 2023). This is driven by practices such as Deep-Sea Tailings Disposal (DSTD), which involves ejecting toxic mining waste directly into the ocean. Such methods have reportedly devastated local marine ecosystems, leading to observations of fish disappearing and the sea turning red in affected regions (Institutional Shareholder Services, 2023). Furthermore, the "opencast" or "strip" mining required to extract the ore has resulted in significant deforestation. One analysis found that in Indonesia alone, this has resulted in the clearing of more than 76,301 hectares of tropical forest (Guberman et al., 2024).

While prior research has characterized the global nickel trade in terms of physical and monetary flows (Reck et al., 2008; Sun et al., 2024), supply risks (Tian et al., 2021), or logistical cost optimization (Dong et al., 2021), a significant gap remains. Less attention has been paid to how the sustainability performance of countries relates to the structure of the global nickel trade network. We know *what* is traded, *how much*, and what the *most efficient* logistical routes are, but we do not fully understand the underlying logic of *who* trades with *whom*, and whether sustainability is a factor in those decisions. This gap is a critical concern; if the green transition is to be truly sustainable, it must account for the environmental impacts along its entire supply chain. Understanding whether sustainability influences trade patterns is crucial for policymakers and industries aiming to mitigate environmental hotspots and build resilient, responsible supply chains.

To investigate this gap, this paper addresses two primary research questions.

RQ1: What characteristics explain how countries trade nickel with each other?

To gauge a country's sustainability, the Environmental Performance Index (EPI) is used, which is a score that ranks 180 countries on climate change performance, environmental health and ecosystem vitality (Block et al., 2024). EPI has been described as a popular sustainability index, widely used in research (Pinar, 2022). Moreover, control variables are included. GDP is included, because

larger economies systematically trade more and occupy more central positions in trade networks (Anderson van Wincoop, 2003; World Bank, 2024). LSCI (Liner Shipping Connectivity Index) is included because it measures how well a country is integrated into global liner shipping networks, reflecting maritime transport connectivity that strongly shapes trade costs and export performance (Fugazza Hoffmann, 2017; UNCTAD, 2025), and geographical distance is included, because it is a standard determinant of trade intensity in gravity models, with trade typically decreasing in distance (Anderson van Wincoop, 2003; Mayer Zignago, 2011).

Based on these theoretical considerations, we formulate the following hypothesis:

H1: Countries with similar EPI tend to trade nickel with each other. *Controlling for GDP, LSCI and geographical distance.*

While the first research question assesses the determinants of trade intensity, the second research question shifts focus to the drivers of trade partnership formation.

RQ2: Why do countries trade nickel?¹

Moreover, to capture how strongly countries are involved in nickel-intensive clean technologies, we include two distinct measures: exports of lithium-ion accumulators (HS 850760) and exports of battery-electric passenger vehicles (HS 870380). Exports of Li-ion accumulators serve as an indicator of upstream battery manufacturing, as modern Li-ion chemistries are heavily nickel-based and these batteries can be used across EV, stationary storage and other applications. In contrast, exports of battery-electric passenger vehicles proxy downstream EV assembly, which embeds large amounts of nickel in traction batteries and reflects countries that export finished clean-technology goods. Using both variables allows us to test whether nickel imports are more closely associated with upstream component production or with downstream exports of finished clean-technology products. Both variables are constructed from bilateral trade data in the UN Comtrade database using these HS codes (United Nations Statistics Division, 2024).

H2: Countries with higher EV exports tend to buy more nickel.

H3: Countries with higher battery exports tend to buy more nickel.

Nickel trade has increasingly become a geopolitical topic. Electrification is not only a sustainability issue, but it has also become a core strategic topic for countries' ability to keep meeting their energy demands, and to be on the frontier of new technologies and markets. Therefore, countries carefully consider their nickel trading partners (Jia et al., 2024; The Netherlands, 2022). Moreover, an analysis of the nickel trade by Ma et al. (2022) shows a persistent core of countries that maintain many trade relationships over time, while most countries trade with only a few partners. This structure is characteristic of "rich-get-richer" or preferential attachment dynamics, where nodes with many ties are more likely to obtain new ties (Barabási Albert, 1999). Based on this research the following hypotheses were formulated:

H4a: Countries that already import from many partners tend to attract even more import partners

H4b: Countries that already export to many partners tend to attract even more export partners

¹In Appendix A.2 the exclusion of hypothesis 5 is explained.

Table 1: Hypotheses, ERGM terms, and motivations.

Hypotheses	ERGM Term	Motivation
H2: Countries with higher EV exports tend to buy more nickel.	nodeicov("ln_EV_exports")	The nodeicov term tests whether an actor's attribute (high EV exports) is correlated with its number of incoming ties (imports).
H3: Countries with higher battery exports tend to buy more nickel.	nodeicov("ln_battery_exports")	The nodeicov term tests whether an actor's attribute (high battery exports) is correlated with its number of incoming ties (imports).
H4a: Countries that already import from many partners tend to attract even more import partners	istar(2) and istar(3)	Both istar(k) terms are included to model the "rich-get-richer" dynamic accurately. The istar(2) term captures basic inequality (testing if some countries are more popular importers than others). The istar(3) term tests for extreme concentration (the existence of "super-importers"). Including both terms allows the model to determine if the network is driven by a few dominant buyers attracting a disproportionate number of partners.
H4b: Countries that already export from many partners tend to attract even more export partners	ostar(2) and ostar(3)	Similarly, both ostar(k) terms are used to verify if the export market is dominated by a small "core" of suppliers. The ostar(2) term measures the general spread between small and large exporters, while the ostar(3) term specifically captures the presence of major export "hubs."

Overall, this paper makes three distinct contributions to the literature on the global nickel trade. First, by combining the EPI with standard gravity controls (GDP, LSCI, and distance) in a network framework, it moves beyond simple volume statistics to determine whether countries with higher sustainability performance occupy distinct positions within the trade network. Second, by relating bilateral nickel flows to national roles in EV and battery export markets, the study links upstream nickel trade to the geography of clean-technology production, distinguishing between countries that demand nickel for the green transition versus those that are merely trade intermediaries. Third, the joint application of MRQAP and ERGM applies modern network methods to uncover not just attribute-based drivers, but also endogenous structural dynamics, specifically testing for "rich-get-richer" centralization effects among importers and exporters. Together, these elements provide a more integrated picture of "who trades nickel, why, and with what environmental implications." The remainder of this paper is organized as follows: Section III describes the dataset and variable operationalization. Section IV outlines the research rationale and dual-method network approach. Section V presents the empirical results for both studies, and Section VI concludes with policy implications.

III. DATASET

For our network analysis, we combine multiple datasets to capture all attributes relevant for the analysis.

I. Variables and Datasources

The network is constructed around 2024 nickel trade. This dataset is collected from the UN Comtrade database by using code HS 2604 for nickel. This dataset captures nickel export flows from countries and reporting economic territories. The UN Comtrade database collects trade data from statistical authorities and composes this into a trade database (UN Statistics Division, n.d.). This data is constructed into a directed weighted network where nodes are reporting countries and edge weights represent the value of the nickel exports in USD. For the ERGM in Study 2 we use a binarised version of this network, where any positive trade value is coded as a tie. We opt for this threshold of zero because the network has a low density score. Therefore, our focus lies on the existing trade ties.

To build on this core trade network, four additional datasets were integrated. Namely, an Environmental Performance Index (EPI) for 2024, which is an indicator that ranks countries on environmental performance. This measure is calculated yearly by researchers from Yale and Columbia universities (Block et al., 2024). The 2024 edition combines 58 indicators across 11 issue categories, including climate change mitigation and waste management. EPI is the primary focus of Study 1.

Economic size is added using 2024 GDP-data, sourced from the World Bank World Development Indicators (World Bank, 2025). The World Bank harmonises values obtained from national statistics agencies and composes this into a consistent international GDP dataset. GDP is used in both study 1 and study 2.

The LSCI is used as an index that captures how well countries are connected to global shipping networks. This data is also composed by the World Bank and is based on 5 components of a country’s maritime sector. The World Bank indicates that around 80% of global trade by volume is shipped via sea, meaning that maritime connectivity is essential for a country’s access to world markets (UNCTAD, 2025). However, LSCI only measures maritime connectivity and leaves out nickel trade via land and sea, which could negatively bias landlocked countries or countries which ship nickel via non-maritime methods. LSCI is used as a control variable for Study 1.

Complementary to shipping connectivity, geographical distance measures geographical separation between countries. This data is obtained from the CEPII database. It measures geographical distance by obtaining a country’s population distribution and measuring the distance between another country’s population distribution (CEPII, 2022). Geographical distance is used as a control variable for Study 1.

For Study 2, two more variables were collected, namely trade data on battery exports and EV exports, also from the UN comtrade database. The codes and more information about these variables are laid out in Appendix B.

Overall, there were some missing data; the handling of this missing data is explained in Appendix B, along with the operationalization of the variables.

II. Descriptive Statistics

The network consists of 78 countries and 213 directed trade ties. It has a density of 0.035, meaning that 3.5% of all possible trade ties are present. The average degree is 2.73, indicating that countries trade with only a small subset of all possible partners. However, the plot shows that the average does not capture the full complexity. Although most countries are located around the y-axis, there are some outliers which have higher degrees, indicating the existence of trade hubs.

Reciprocity is 0.197, suggesting that 19.7% of export ties are reciprocated by an import. The transitivity of 0.204 indicates a modest tendency towards triadic closure, where trade partners of a country are also connected among each other. Additional descriptive plots are included in the appendix.

Moreover, the node centrality metrics reveal a highly unequal and fragmented network characterized by a distinct core-periphery structure. The heavy right-skew in degree and betweenness centrality reveals that while most nations have minimal connectivity, a select few "super-hubs" dominate global flows and act as critical bridges. This configuration points to a fragile supply chain heavily reliant on a small number of central actors.

Statistic	Measure
Vertex Count	78
Edge Count	213
Density	0.035
Reciprocity	0.197
Transitivity	0.204
Number of Isolates	0

Table 2: Basic Network Descriptive Statistics

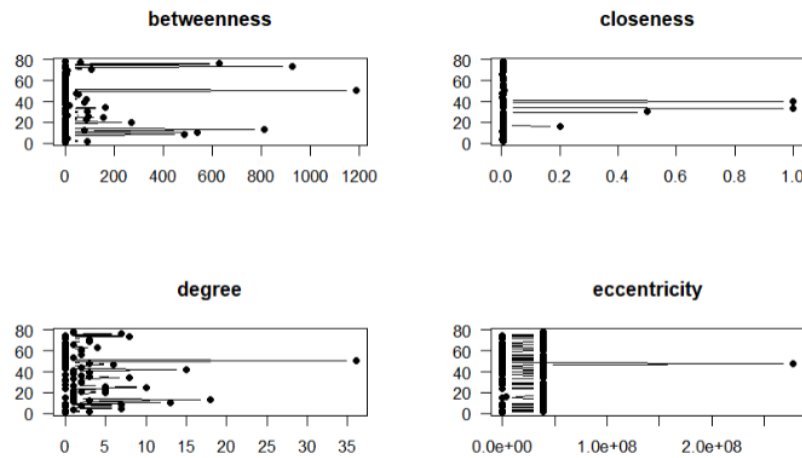


Figure 1: Distribution of Node Centrality Metrics

IV. RESEARCH RATIONALE

This project combines two complementary network-analytic approaches, MRQAP and ERGM, to study both attribute-based and structural factors within global nickel trade. The first study applies a Multiple Regression Quadratic Assignment Procedure (MRQAP) to test whether trade intensity between countries is related to their similarity in environmental performance. MRQAP is particularly well suited for dyadic trade data because it corrects for the autocorrelation among network ties that violates the independence assumption in traditional OLS regression (Krackhardt, 1988). By regressing bilateral nickel trade intensity on sustainability similarity (EPI), while controlling for GDP, maritime connectivity (LSCI) and geographical distance, MRQAP isolates whether sustainability alignment explains trade relations beyond conventional gravity forces. This method is particularly appropriate for continuous and directed trade networks, as it captures how similarities between countries translate into trading patterns. In practice, the MRQAP permutation framework preserves the row/column dependence structure of the trade matrix, yielding valid tests even when dyads are not independent.

The second study uses an Exponential Random Graph Model (ERGM) to examine the structural dependencies that underlie the organization of the nickel trade network. ERGMs are explicitly designed for interdependent tie formation and allow the joint modelling of exogenous node-level attributes and endogenous structural effects (Robins, 2007; Lusher, 2013; Hunter, 2008). This enables us to test whether countries that are more engaged in clean-technology production that is measured through their exports of electric vehicles and lithium-ion batteries, also tend to import larger amounts of nickel, while at the same time accounting for degree-based concentration in the network. Including these variables helps connect the trade of raw materials like nickel to countries' involvement in the clean-technology value chain, showing how economic interests and environmental objectives interact in the same network.

Traditional econometric models, including standard OLS dyadic regressions and the gravity model, struggle with two core challenges of network data. First, they typically assume dyadic independence conditional on covariates, so they cannot represent reciprocity, clustering or degree concentration that arise from interdependent tie formation (Robins, 2007). Second, in international trade specifically, the gravity framework is the standard model but ignores network dependencies among bilateral flows, hence it is often empirically inadequate when such dependencies matter (Ward, Ahlquist, Rozenas, 2013). These limitations motivate the use of ERGMs, which explicitly parameterise structural dependence and can reproduce complex network configurations.

MRQAP and ERGM address these limitations in complementary ways. MRQAP focuses on who trades nickel with whom and whether sustainability alignment matters, while delivering valid inference under dyadic dependence (Krackhardt, 1988). ERGM, in turn, models the formation of trade ties as a function of both exogenous attributes and endogenous mechanisms, enabling tests about how network organization and industrial demand shape the pattern of trade (Robins et al., 2007; Lusher et al., 2013). Together, the two approaches offer a clear and rigorous picture of the nickel trade network, combining sustainability performance, economic connectivity and structural dynamics in a single analytical framework.

V. RESULTS

I. Study 1

Our first research question asks which characteristics explain how countries trade nickel with each other, and we hypothesise that pairs of countries with more similar Environmental Performance Index (EPI) scores trade larger volumes of nickel with each other, controlling for economic size, maritime connectivity and geographic distance.

To answer RQ1, we estimate a Multiple Regression Quadratic Assignment Procedure (MRQAP) model in which the dependent variable is the natural logarithm of bilateral nickel exports plus one, $\log(\text{trade}_{ij} + 1)$. The main predictor is environmental similarity, operationalised as the absolute difference in EPI scores between country i and j ($|EPI_i - EPI_j|$). Here, smaller values indicate more similar environmental performance. We control for 3 variables. Economic size is captured through the logarithm of the product of exporter and importer GDP ($\log(GDP_i \times GDP_j)$). Maritime connectivity is captured by the sum of exporter and importer LSCI scores. Geographical distance is captured by the logarithm of population-weighted geographic distance ($\log(\text{distance}_{ij} + 1)$). The rationale behind these operationalizations is included in Appendix C.1. The model is estimated using MRQAP with 1,000 permutations, which preserves the row and column dependence in the trade matrix and yields valid permutation-based p-values for each coefficient.

To assess how well the MRQAP captures variation in bilateral nickel trade, we computed dyad-level model fit statistics and examined permutation and residual diagnostics. The model explains a modest share of the variance in log nickel exports ($R^2 = 0.06$; adj. $R^2 = 0.06$), with a correlation of about 0.25 between observed and fitted dyadic values and small average prediction errors (RMSE ≈ 0.12 , MAE ≈ 0.05 ; model fit statistics table is included in Appendix C.2). Permutation histograms show that the observed coefficients lie in the expected tails of their permutation distributions (Appendix C.3). Given the sparsity and skewness of the trade network, this level of fit is reasonable and in line with our goals.

```

--- Running MRQAP (Clean) ---

OLS Network Model

Residuals:
      0%      25%      50%      75%
-0.1928923485 -0.0386457210 -0.0183204780  0.0004840434
      100%
 0.7097109355

Coefficients:
              Estimate      Pr(<=b) Pr(>=b) Pr(>=|b|)
(intercept) -1.581710e-01 0.097    0.903    0.207
x1          -5.080757e-04 0.042    0.958    0.086
x2           4.531439e-03 0.970    0.030    0.057
x3           8.224028e-05 0.996    0.004    0.004
x4          -8.732302e-03 0.005    0.995    0.008

Residual standard error: 0.1242 on 6001 degrees of freedom
Multiple R-squared: 0.06223    Adjusted R-squared: 0.06161
F-statistic: 99.56 on 4 and 6001 degrees of freedom, p-value:

```

Figure 2: MRQAP results

The table summarizes the MRQAP results for bilateral nickel trade. The coefficient of environmental similarity is negative, as hypothesized, but not significant ($\beta \approx -0.0005$, $p = 0.086$). The pattern points in the expected direction: country pairs with more similar EPI scores tend to trade more nickel, but the statistical evidence is weak and does not come close to the 5% significance threshold.

The gravity-type controls behave broadly in line with expectations. Economic size has a positive, borderline significant effect ($\beta \approx 0.0045$, $p = 0.057$), indicating that larger economies trade more nickel with each other. Maritime connectivity shows a seemingly small, but statistically significant effect ($\beta \approx 0.00008$, $p = 0.004$), suggesting that linear shipping connectivity has a positive effect on nickel trade. The coefficient's small magnitude results from the high scale of LSCI scores. Moreover, geographic distance is clearly important: its coefficient is negative and statistically significant ($\beta = 0.0087$, $p = 0.008$), consistent with the classic gravity result that more distant countries trade systematically less.

Taken together, these findings indicate that traditional gravity forces, mainly distance and to a lesser extent economic size, remain the main drivers of bilateral nickel trade in 2024, while environmental similarity plays at most a secondary role. In other words, H1 is not supported empirically and we fail to reject the null hypothesis. Although the sign of the effect is consistent with the hypothesis, the significance is limited once standard trade determinants are controlled for. This suggests that sustainability choices do not leave a strong imprint on aggregate trade volumes. Study 2 further expands on what factors play a role in the nickel trade network, which products (EVs and batteries) drive nickel trade and which structural factors exist in the network.

II. Study 2

II.1 Model Specification and Selection

To address the second research question concerning the drivers of the global nickel trade, we utilized Exponential Random Graph Models (ERGMs) to test the hypothesized effects of exogenous economic attributes and endogenous network structures. Specifically, this analysis evaluates whether higher EV and battery exports predict increased nickel purchasing (H2 and H3) using nodecov terms, and assesses the presence of preferential attachment, where popular traders attract disproportionately more partners, using istar and ostar terms (H4a and H4b).

To execute this analysis, we estimated a series of nested models. We began with a baseline model (Model 1) containing only the edges term. Model 2 introduced the main exogenous predictors related to our hypotheses: nodecov (incoming ties) for `ln_ev_exports` and `ln_batt_exports`. We applied a natural logarithmic transformation to these variables to standardize their highly skewed distributions, thereby reducing the influence of extreme outliers and improving model stability (see Appendix D.2).

To improve model fit, we tested the inclusion of Gross Domestic Product (GDP) as a control variable in Model 3, adding both nodecov (receiver effect) and nodecov (sender effect) for `ln_gdp`, also standardizing for model stability (Appendix D.2). A comparison of the model fit statistics reveals a substantial improvement from Model 2 (AIC = 1809.75; BIC = 1829.86) to Model 3 (AIC = 1587.30; BIC = 1620.81). The lower AIC and BIC values indicate that including economic size (GDP) significantly explains the network structure better than the hypothesis variables alone; therefore, GDP was retained as a control in subsequent models.

Finally, we estimated two separate structural models to test for preferential attachment (Hypotheses 4a and 4b) (see Appendix D1 for further explanation). Model 4a focuses on import dynamics by

including $\text{istar}(2)$ and $\text{istar}(3)$ terms, while Model 4b focuses on export dynamics by including $\text{ostar}(2)$ and $\text{ostar}(3)$ terms. These represent our final models for analysis.

II.2 MCMC Diagnostics and Goodness of Fit (GOF)

Based on the Goodness-of-Fit (GOF) comparison, Model 4b provides a more realistic representation of the higher-order structure of the nickel trade network than Model 4a. In particular, Model 4b substantially improves the fit for edgewise shared partners and geodesic distance distributions, indicating a better capture of clustering and global connectivity patterns.

In addition, Model 4b achieves a substantially lower AIC (1439.04) and BIC (1485.94) compared to Model 4a (AIC: 1559.37; BIC: 1606.27), suggesting superior explanatory power. For these reasons, Model 4b is used as the primary reference for interpreting network structure and hypothesis-related mechanisms.

At the same time, MCMC diagnostics indicate remaining estimation challenges, particularly for higher-order dependence terms. Therefore, hypothesis testing emphasizes the direction and robustness of estimated effects rather than relying exclusively on statistical significance. Detailed diagnostics and convergence checks are reported in Appendix D.3.

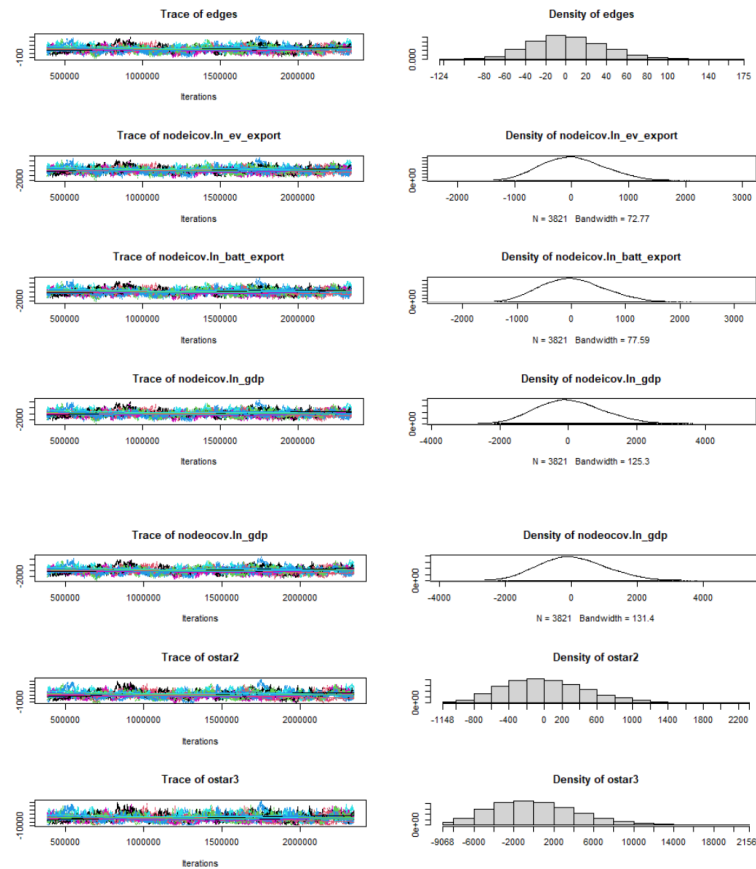


Figure 3: MCMC Diagnostics for Model 4b

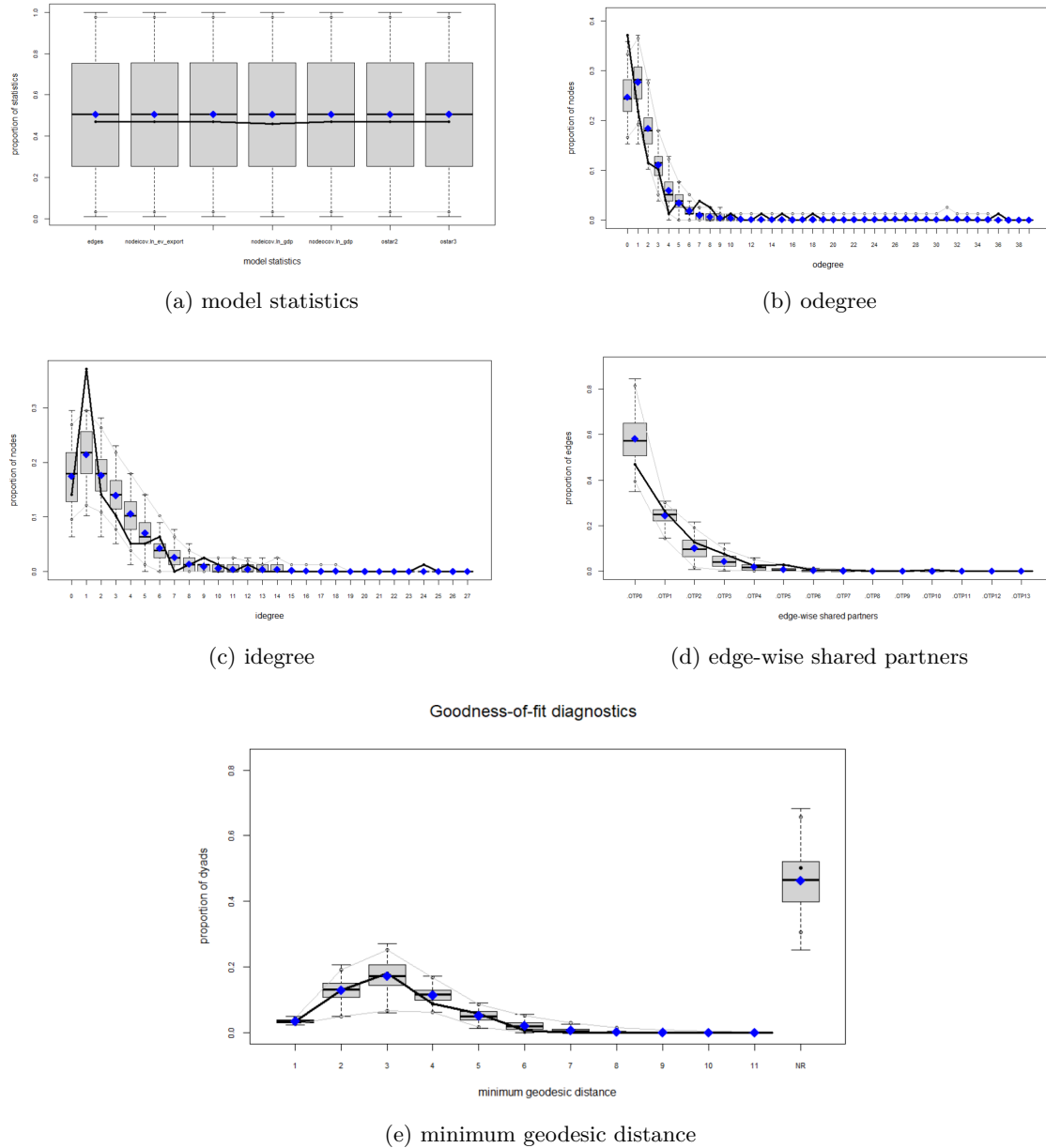


Figure 4: Goodness-of-fit (GOF) Diagnostics for Model 4b.

II.3 ERGM Results

Exogenous Hypotheses (H2 and H3)

Hypothesis 2 posited that countries with higher EV exports tend to import more nickel. The results from Model 4b do not support this prediction in the expected direction. The coefficient for $\ln_ev_export$ is negative (0.06) and statistically significant ($p < 0.01$). The odds ratio (0.94) implies that for every one-unit increase in the log of EV exports, the odds of a country importing nickel actually decrease by approximately 6%, holding all other structures constant. Therefore,

hypothesis 2 is rejected.

Hypothesis 3 predicted that countries with higher battery exports would buy more nickel. The results support this hypothesis. The coefficient for `nodecov.ln_batt_export` is positive (0.06) and significant ($p < 0.05$). This yields an odds ratio of 1.06, indicating that a one-unit increase in the log of battery exports is associated with a 6% increase in the odds of forming a nickel import tie. Therefore, hypothesis 3 is supported.

Endogenous Hypotheses (H4a and H4b)

Hypothesis 4a proposed a “rich-get-richer” dynamic for importers (preferential attachment). Although the `istar` terms in Model 4a were significant (e.g., `istar(2)` was 0.20, $p < 0.001$) the model failed to adequately capture the global network structure, as established by the poor Goodness of Fit (see Appendix D3 for further explanation). Therefore, hypothesis 4a is not supported due to model misspecification.

Hypothesis 4b predicted that countries already exporting to many partners tend to attract even more export partners. This was tested in Model 4b and is strongly supported. The positive and significant `ostar(2)` coefficient (0.26, $p < 0.001$) indicates a “snowball effect” where exporting is essentially contagious: once a country establishes two export ties, the odds of them forming subsequent ties increase by approximately 30% (Odds Ratio: 1.30). This confirms the market relies on a core of dominant suppliers. However, the model also identifies a natural limit to this expansion. The `ostar(3)` term is negative and significant (-0.01, $p < 0.001$), acting as a “brake” or damping effect. This ensures the model does not predict unrealistic “super-hubs” that monopolize the entire market; while major hubs exist, their dominance has a ceiling. Therefore, hypothesis 4b is supported.

Control variables

GDP remained a consistent driver of trade. In Model 4b, the effect of GDP on importing is large and positive (coeff: 0.55, $p < 0.001$), with an odds ratio of 1.73. Conversely, the effect of GDP on exporting is much smaller (coeff: 0.08, $p < 0.01$, OR: 1.08). This suggests that while economic size dictates import capacity, the “hub” structure of the network mediates much of the direct influence of economic size on exports (see Appendix D4 for a detailed analysis of this structural mediation).

	Model 1	Model 2	Model 3	Model 4a	Model 4b
edges	-3.30 *** (0.07)	-4.54 *** (0.26)	-30.41 *** (1.92)	-24.06 *** (1.92)	-21.42 *** (1.65)
nodeicov.ln_ev_export		0.00 (0.02)	-0.06 * (0.02)	-0.03 (0.02)	-0.06 ** (0.02)
nodeicov.ln_batt_export		0.07 ** (0.02)	0.06 * (0.03)	0.03 (0.02)	0.06 * (0.03)
nodeicov.ln_gdp			0.48 *** (0.05)	0.20 *** (0.05)	0.55 *** (0.06)
nodeocov.ln_gdp			0.52 *** (0.04)	0.54 *** (0.05)	0.08 ** (0.03)
istar2				0.20 *** (0.04)	
istar3				-0.01 * (0.00)	
ostar2					0.26 *** (0.02)
ostar3					-0.01 *** (0.00)
AIC	1842.86	1809.75	1587.30	1559.84	1439.04
BIC	1849.56	1829.86	1620.81	1606.74	1485.94
Log Likelihood	-920.43	-901.88	-788.65	-772.92	-712.52

*** p < 0.001; ** p < 0.01; * p < 0.05

Figure 5: Results ERGM Models

VI. CONCLUSION

This study investigated the structure of the 2024 global nickel trade network to determine if environmental performance predicts trade partnerships (Study 1) and whether leadership in clean-technology exports drives nickel demand (Study 2).

Study 1 addressed the research question: *Which characteristics explain how countries trade nickel with each other, and do countries with more similar environmental performance trade larger volumes of nickel?* Using a valued MRQAP model, we examined whether environmental similarity predicts nickel trade, controlling for gravity-type controls (economic size, maritime connectivity and geographic distance). The results show that traditional trade determinants remain the primary drivers of nickel trade: geographic distance is strongly and negatively associated with trade volumes, and economic size exerts a weaker but positive effect, while maritime connectivity does not add much explanatory power once these factors are controlled for. Environmental similarity behaves as hypothesised in terms of sign, but its effect is small and only marginally significant, offering at best tentative support for H1. In line with the modest model fit statistics, these findings suggest that sustainability alignment leaves only a faint imprint on aggregate nickel trade volumes. This indicates that nickel trade is mainly driven by other factors rather than sustainability goals.

Study 2 addressed the research question: *Why do countries trade nickel?* To answer this, four specific hypotheses were formulated regarding the drivers of trade formation. Hypothesis 2 (H2) and Hypothesis 3 (H3) examined industrial demand, predicting that countries with higher exports of Electric Vehicles (EVs) and batteries, respectively, would be more likely to import nickel. Hypotheses 4a and 4b investigated structural preferential attachment, proposing that current high-volume importers (H4a) and exporters (H4b) would attract even more partners, following a "rich-get-richer" dynamic.

The analyses yielded three critical insights into these drivers and the overall structure of the global nickel trade network.

First, the results indicate that the network is structurally defined by export centralization rather than import centralization. The superior fit of Model 4b over Model 4a indicates that the trade system is primarily driven by a "core" of dominant suppliers (exporters) who benefit from preferential attachment dynamics, effectively acting as hubs. Conversely, the "rich-get-richer" dynamic does not apply to importers; there is little evidence for equivalent "super-importers" structuring the network.

Second, regarding industrial drivers, the analyses revealed a distinct functional separation between battery production and EV production. While battery export capacity was a significant predictor of nickel imports (supporting H3), EV export capacity was inversely related (rejecting H2). This contrast is consistent with the emergence of a middle processing hub: countries that dominate final EV assembly appear to bypass raw nickel sourcing, relying instead on these intermediary hubs to import raw materials and export processed battery components.

Third, while economic size (GDP) remains a powerful baseline driver for trade formation, its influence is partly mediated by the network's structural constraints. The analyses showed that once the tendency for export centralization is accounted for, the marginal effect of GDP on exporting drops significantly.

These findings are particularly valuable for policymakers who are concerned with both environmental issues and geopolitical strategy involving nickel trade as a critical mineral. Additionally, supply chain managers in the EV industry benefit by understanding that nickel sourcing currently follows traditional gravity and cost-based logic rather than sustainability alignment, signaling a risk of

'greenwashing' in upstream supply chains. Finally, Non-Governmental Organizations focused on sustainability can use this evidence to push for stricter transparency, showing that the 'green' transition currently relies on trade patterns indistinguishable from conventional dirty industries.

While a significant open problem is the causal direction of the relationship between trade and sustainability, the cross-sectional approach used reveals that green countries do not preferentially trade with each other, but it cannot determine if trading with lower-standard partners actively degrades a country's EPI score over time or if it simply reflects static natural resources. Furthermore, the analysis is limited by the opacity of domestic consumption and intermediate processing, often referred to as the 'black box' of mineral supply chains (Mancheri et al., 2019). Our findings in Study 2 empirically confirm this 'black box' phenomenon: the lack of a significant relationship between EV exports and nickel imports demonstrates that the final producers of green technology are structurally disconnected from raw material sourcing, relying entirely on opaque intermediary processing hubs. Future research could employ Temporal Exponential Random Graph Models (TERGM) to analyze network evolution over time, specifically before and after the implementation of major green industrial policies (e.g., 2020 vs. 2025) (Leifeld et al., 2018). Additionally, a multiplex network analysis combining nickel trade with other critical battery minerals (lithium, cobalt, graphite) would provide a more holistic view of the 'green transition' supply chain (Kivelä et al., 2014). In addition, incorporating firm-level data rather than just country-level aggregates would allow for a more granular assessment of whether specific multinational corporations are successfully aligning their procurement strategies with their sustainability pledges. Furthermore, future simulations could assess network resilience against geopolitical shocks. By modeling the hypothetical removal of central nodes (e.g., Indonesia, Russia) researchers could quantify the structural fragility of the green transition supply chain. Additionally, incorporating geopolitical alliance data (e.g., UN voting patterns) as a covariate would help determine if the trade network is fragmenting into opposing political blocks, revealing whether 'friend-shoring' strategies are effectively reshaping the global nickel market alongside sustainability goals.

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VIII. APPENDIX

I. Appendix A

I.1 A.1 Github Link

Here is the link to our Github repository that contains the code and datasets used for this assignment:
<https://github.com/graveyarddata/SNA/tree/main>

I.2 A.2 Exclusion of Hypothesis 5 (Reselling Dynamics)

Nickel is strategically important and countries deliberately shape their nickel trading patterns in response to geopolitical circumstances. Consequently, there is room for intermediaries that import nickel from country X and resell it to country Y, even when X and Y would not trade directly. This kind of role is consistent with the findings of Zhou et al. (2023), who identify transmission hubs in the global nickel trade. Normally, raw materials such as nickel would flow directly from low-cost producers to end users, so there is little economic reason for countries to systematically buy and resell the same material without adding value, consistent with arbitrage literature (Richardson, 1978). This led to the following hypothesis:

H5: There are countries that buy and resell nickel.

To test this, we formulated the specific hypothesis and corresponding ERGM term as shown in Table B.1 below.

Table 3: Hypothesis 5 Specification

Hypotheses	ERGM Term	Motivation
H5: There are countries that buy and resell nickel.	twopath	The twopath term counts the frequency of (A $\rightarrow$ B $\rightarrow$ C) structures, capturing the dynamic of "buy-resell" activity by the middle actor (B).

Convergence Issues for H5: We encountered persistent convergence issues when attempting to estimate models including the twopath statistic. To rigorously test if the model could be stabilized, we experimented with multiple specifications:

1. Baseline Models: We tested the twopath term in isolation (with only the edges term) to see if the structural effect itself was causing instability.
2. Covariate Models: We attempted to include twopath alongside node covariates such as GDP, EV exports, and Battery exports to control for exogenous effects.
3. Structural Combinations: We tried combining twopath with other structural terms, specifically trying it in conjunction with degree-based terms (like idegree or odegree) and shared partner statistics (gwesp).

In all cases, the Markov Chain Monte Carlo (MCMC) estimation failed to converge. The exclusion of Hypothesis 5 was necessitated by the well-documented statistical instability of the twopath term in Exponential Random Graph Models. As noted by Snijders et al. (2006), terms that model triadic or path-based closure (such as twopath or simple transitivity) without geometric weighting are

prone to 'degeneracy.' This occurs because these terms create a cascading effect: adding a single edge creates multiple new paths, which increases the probability of adding more edges, leading the simulation to runaway into a completely dense or 'full' graph that no longer represents the observed data. Consequently, the model could not converge, and the hypothesis was dropped.

II. Appendix B

B.1: Missing data handling

Missing data is handled in a structured way. We harmonise country names across all sources using ISO3 codes and an extensive synonym list. For remaining gaps, we apply the following rules: 1) Landlocked countries that do not have seaports are assigned an LSCI value of zero, 2) Important economies with missing values in specific sources (e.g., New Caledonia in the GDP file) are manually completed using World Bank data from 2022 (which was the last credible reporting source), 3) EV and battery exports that are missing are treated as zero and 4) Any residual missing values for EPI, GDP and LSCI among very small territories are imputed using the sample mean.

B.2: Battery exports and EV exports

Both variables are derived from the same UN Comtrade system as the nickel data. EV exports are extracted using HS code 870380, which corresponds to “vehicles with only electric motor for propulsion” (United Nations Statistics Division, n.d.), while battery exports are extracted using HS code 850760, defined as “electric accumulators; lithium-ion, including separators, whether or not rectangular (including square)” (United Nations Statistics Division, n.d.). For each reporter, we aggregate annual export values across all partners to obtain total EV and lithium-ion battery exports in USD, which are then attached as node attributes. They are log-transformed in the analysis to reduce skewness and are included in Study 2 to test whether such involvement is associated with structurally different positions in the global nickel trade network.

B.3: Descriptive statistics of network

`$number_of_vertices`

[1] 78

`$number_of_edges`

[1] 213

`$density`

[1] 0.035

`$reciprocity`

[1] 0.197

`$transitivity`

[1] 0.204

`$mean_distance`

[1] 5763089

`$number_of_isolates`

[1] 0

`$dyad_census`

	Mutual	Asymmetric	Null
1	21	171	2811

`$triad_census`

	003	012	102	021D	021U	021C	111D	111U	030T	030C	201	120D	120U	120C	210
300															
1	63718	9201	1086	762	305	442	141	232	89	9	26	15	20	12	16
2															

III. Appendix C: Study 1

III.1 C.1: Operationalisation of MRQAP variables

This appendix provides additional detail on how the variables in the MRQAP model for RQ1 are constructed, why we use these specific operationalisations, and which alternative specifications would be reasonable.

C.1.1 Nickel trade

The dependent variable is a dyadic matrix of bilateral nickel exports in 2024. Starting from the weighted adjacency matrix of trade flows in USD, we apply a log-plus-one transformation:

$$Y_{ij} = \log(\text{trade}_{ij} + 1).$$

This transformation addresses the extreme right-skewness of trade values and reduces the influence of a small number of very large flows. It is also consistent with gravity models, where bilateral trade is theorised to follow a multiplicative structure in economic size and trade costs; taking logs linearises this structure and makes coefficients more easily interpretable (Anderson & Van Wincoop, 2003; Ward et al., 2013).

A natural alternative would be to model a binary outcome (presence/absence of trade) rather than volumes. That would emphasize ‘who trades with whom’ instead of ‘who trades how much’. We opted for the log volume specification because RQ1 explicitly asks which characteristics explain how much countries trade nickel with each other, not just whether a tie exists.

C.1.2 EPI

For each pair of countries we compute the absolute difference in their EPI-scores:

$$\text{EPI_diff}_{ij} = |\text{EPI}_i - \text{EPI}_j|$$

This yields a dyadic “distance” on the EPI scale. Smaller values indicate more similar environmental performance. A negative coefficient on this term indicates that more similar countries trade more nickel, conditional on controls. However, it is plausible that environmental similarity is non-linear and, for example, only happens at the high end of the EPI scale. This would indicate that EPI similarity only matters for countries really focusing on sustainability. This could be modelled with interaction terms (e.g. $(\text{EPI_diff} \times \text{mean}(\text{EPI}_i, \text{EPI}_j))$).

C.1.3 GDP Economic size is based on GDP in USD. To capture the economic size of a trade relation, we construct the product of exporter and importer GDP and then take logs:

$$\text{GDP_prod}_{ij} = \log(\text{GDP}_i \times \text{GDP}_j)$$

The GDP product is a multiplicative measure of economic size: it reflects the idea from structural gravity models that bilateral trade grows with the product of the two partners’ GDPs (Anderson & van Wincoop, 2003). Taking the logarithm of this product is essentially the same as including log GDP for exporter and importer separately, but in a more compact dyadic term, and it keeps our specification in line with the standard log-linear gravity framework. Instead of $\log(\text{GDP}_i \times \text{GDP}_j)$, one could include $\log(\text{GDP}_i)$ and $\log(\text{GDP}_j)$ as two separate covariates. This would allow testing

whether exporter and importer size have asymmetric effects (e.g. large importers vs large exporters). Moreover, to study whether highly asymmetric pairs (large-small) trade differently from balanced pairs (large-large or small-small), one could add a term such as $|\log(\text{GDP}_i) - \log(\text{GDP}_j)|$. That would shift the focus from “how big is the pair?” to “how unequal is the pair?”.

C.1.4 Maritime connectivity (LSCI sum)

Maritime connectivity is measured with the Liner Shipping Connectivity Index (LSCI). For each dyad we sum the LSCI scores of the two countries:

$$\text{LSCI_sum}_{ij} = \text{LSCI}_i + \text{LSCI}_j$$

This specification treats the combined connectivity of both partners as relevant for their ability to exchange goods by sea: higher connectivity at either end is assumed to reduce frictions, and the additive form keeps the model simple and linear. Alternatively, the minimum or maximum could have been used in order to focus on either bottlenecks with respect to maritime connectivity or maritime hubs.

Minimum: $\text{LSCI_min}_{ij} = \min(\text{LSCI}_i, \text{LSCI}_j)$. This reflects a “weakest link” view: even if one country is highly connected, the partner with poor connectivity may limit the overall ease of shipping. This would be appropriate if nickel flows are constrained mainly by access on the “worst” side.

Hub specification (maximum): $\text{LSCI_max}_{ij} = \max(\text{LSCI}_i, \text{LSCI}_j)$. This emphasises the presence of at least one well-connected hub in the pair. It fits a mechanism where connecting to a single major hub (e.g. via transshipment) is sufficient to integrate into the nickel trade network. Otherwise, product could have also been used: $\text{LSCI_prod}_{ij} = \log(\text{LSCI}_i \times \text{LSCI}_j + 1)$; this mirrors the GDP product logic, and emphasises cases where both partners have high connectivity, but it can be numerically unstable when scores are very small or zero.

C.1.5 Geographic distance

Geographic separation is based on CEPII’s population-weighted bilateral distance. For each country pair we take the logarithm of distance in kilometres plus one:

$$\log_dist_{ij} = \log(\text{dist}_{ij} + 1)$$

The log transformation is standard in empirical gravity models: it compresses very large distances, allows for a diminishing marginal effect of distance, and avoids undefined values at very small distances by adding one kilometre. Raw distance could be used, but would give disproportionate influence to the longest-distance pairs and break with the log-linear gravity framework. Additional trade-cost proxies (e.g. common border dummy, common language, regional trade agreements) are widely used in gravity models, but they would make the network more complex.

```

--- MRQAP dyad-level GOF statistics ---
      R2      adjR2      RMSE      MAE cor_y_hat
    0.062     0.062     0.124     0.050     0.249

```

Figure 6: Model Fit Statistics for MRQAP

III.2 C.2: Model Fit Statistics for MRQAP

The table reports regression-based model fit statistics for the MRQAP in Study 1. The model explains about 6% of the variance in log bilateral nickel exports ($R^2 = 0.06$; adj. $R^2 = 0.06$), with relatively small average prediction errors ($RMSE \approx 0.12$, $MAE \approx 0.05$) and a modest correlation between observed and fitted dyadic values ($\text{cor}(y, \hat{y}) \approx 0.25$).

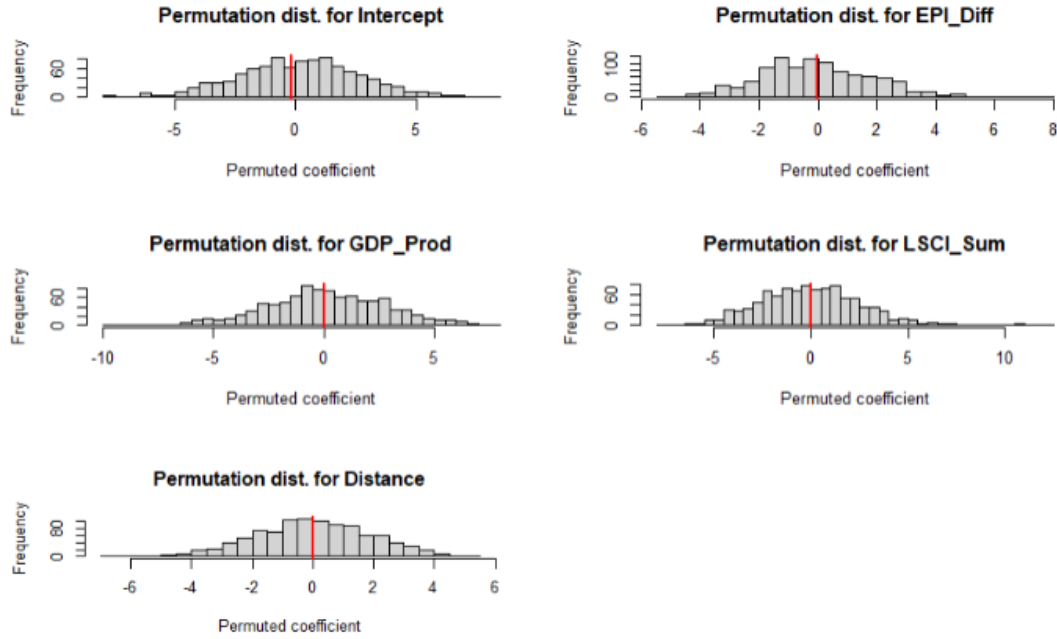


Figure 7: Permutation Distributions of MRQAP coefficients

III.3 C.3: Permutation distributions of MRQAP coefficients.

Histograms show the permutation distributions of the intercept and each dyadic covariate (EPI difference, GDP product, LSCI sum and distance) based on 1,000 MRQAP permutations. Grey bars represent the coefficients obtained under the null hypothesis of no association, while the red vertical lines indicate the observed coefficient estimates from the fitted model. The position of the red lines within these distributions corresponds to the permutation-based p-values reported in Table A2.

IV. Appendix D: Study 2

IV.1 D.1 Separation of Import and Export Preferential Attachment

Our analysis aimed to capture "rich-get-richer" dynamics for both importers (H4a) and exporters (H4b) simultaneously. Theoretically, this would require a single model including both:

- In-star terms (istar(2), istar(3)): To model preferential attachment for importers.
- Out-star terms (ostar(2), ostar(3)): To model preferential attachment for exporters.

In practice, fitting a joint model containing both istar and ostar terms proved computationally infeasible due to severe convergence failures. We attempted multiple configurations to stabilize the estimation, specifically testing cross-combinations of terms to see if lower-order stars could coexist with higher-order stars. The specific combinations we tested included:

- Symmetric High-Order Terms: Testing istar(3) and ostar(3) together.
- Symmetric Low-Order Terms: Testing istar(2) and ostar(2) together.
- Asymmetric Combinations: Testing mixed variations to capture different levels of centralization:
 - istar(3) combined with ostar(2)
 - istar(2) combined with ostar(3)

The decision to separate the analysis was driven by the collinearity and convergence challenges inherent in simultaneously modeling in-degree and out-degree distributions using simple star terms. As described by Robins et al. (2007), including multiple structural terms that model degree distribution skewness can lead to unstable likelihood landscapes, particularly in sparse networks. When both istar and ostar terms are included, they compete to explain the same set of edges, often resulting in poor MCMC mixing where the sampler fails to distinguish whether tie formation is driven by preferential attachment in importing or exporting. By separating these effects, we isolate the specific structural driver (export centralization) without the statistical noise introduced by the competing term:

1. Model 4a: Focuses exclusively on import dynamics (including istar terms).
2. Model 4b: Focuses exclusively on export dynamics (including ostar terms).

IV.2 D.2 Operationalisation of ERGM covariates

This appendix provides a concise clarification of the node-level covariates used in the ERGM analysis for RQ2. The theoretical motivation for logarithmic transformations and gravity-style specifications is already developed in Appendix C (in particular Sections C.3.1 and C.3.3) and is therefore not repeated here. The focus below is on how these principles are adapted to a directed network (ERGM) setting.

D.2.1 EV exports and battery exports

To test whether downstream clean-technology production predicts upstream nickel imports (H2 and H3), country-level exports of electric vehicles and lithium-ion batteries are included as receiver effects using `nodeicov`. These terms capture whether countries with higher export capacity in these sectors are more likely to receive incoming nickel trade ties.

Both variables enter the model in natural logarithmic form:

$$\ln_ev_exports_i = \log(EV_exports_i), \quad \ln_batt_exports_i = \log(Battery_exports_i)$$

The use of logarithms follows the same rationale as for trade volumes discussed in Appendix C.3.1: it addresses extreme right-skewness, allows for diminishing marginal effects, and prevents a small number of dominant exporters from disproportionately influencing the estimates. In the ERGM context, this also improves numerical stability and interpretability of node-level effects.

Alternative specifications, such as untransformed export volumes or rank-based measures, would emphasise absolute scale or relative position rather than proportional differences and were therefore not adopted.

D.2.2 Gross Domestic Product (GDP)

Economic size is measured using GDP in USD and included both as a receiver effect (`nodeicov`) and a sender effect (`nodeocov`). This allows GDP to influence the likelihood of countries acting as major nickel importers and/or exporters.

GDP is log-transformed:

$$\ln_gdp_i = \log(GDP_i)$$

This specification parallels the gravity-based treatment of economic size discussed in Appendix C.3.3, while adapting it to a directed network framework. Including both incoming and outgoing GDP effects allows for asymmetric roles of economic size that would be obscured in a dyadic product term.

Alternative formulations, such as GDP per capita or categorical size classes, would capture different mechanisms (development level or threshold effects) and were therefore not pursued in the main analysis.

IV.3 D.3: MCMC Diagnostics and Convergence Checks

Appendix D.3: MCMC Diagnostics and Convergence Checks

We evaluated convergence and model adequacy for Models 4a and 4b using MCMC trace plots, autocorrelation diagnostics, Geweke diagnostics, and Goodness-of-Fit (GOF) tests.

Model 4b (Selected Model)

The MCMC trace plots for Model 4b show stable behavior across parameters, with chains fluctuating around constant means for edges, node-level covariates, and the structural terms *ostar*(2) and *ostar*(3). No systematic drifting patterns are observed, suggesting that the estimation process reached a stable region of the parameter space.

Autocorrelation remains relatively high for several parameters, particularly the *ostar* terms, with correlations above 0.8 persisting at larger lags. This indicates slow mixing, which is common in ERGMs that include higher-order dependence terms and strongly correlated covariates.

Geweke diagnostics show mixed but acceptable results. While several chains display individual p-values below conventional thresholds—especially for *ostar*(3)—most chains do not exhibit consistent evidence of non-convergence. These results indicate that convergence is sufficient for substantive interpretation, though not ideal in a strict statistical sense.

The GOF diagnostics show that Model 4b reproduces the main structural features of the observed network reasonably well.

Model-level statistics (edges, node covariates, and *ostar* terms) have Monte Carlo p-values close to 1, indicating that the observed values lie well within the simulated distributions.

Geodesic distance distributions are generally well matched, with MC p-values above 0.36 for all distances except infinite paths, which are expected to be difficult to model due to the network's bipartite structure.

Out-degree distributions are substantially better captured than in Model 4a. While the zero out-degree category remains poorly modeled (MC p-value = 0.00), most positive out-degrees fall within the simulated confidence intervals.

Edgewise shared partner statistics are mostly well reproduced, although extreme categories (e.g., high shared-partner counts) show some remaining discrepancies.

Overall, Model 4b provides an adequate representation of the observed network structure for the purposes of this study, despite remaining challenges related to higher-order dependence and slow mixing.

Model 4a (Rejected Model)

Model 4a shows acceptable trace plot behavior, with no strong evidence of drifting or instability. However, the GOF diagnostics reveal clear shortcomings.

Most notably, Model 4a performs poorly on the out-degree distribution, with several degree categories showing Monte Carlo p-values close to 0.00. This indicates that the observed network lies outside the range generated by the model. In addition, multiple edgewise shared partner categories are poorly matched, suggesting that clustering patterns are not adequately captured.

These deficiencies indicate that the *istar*-based specification in Model 4a does not sufficiently model export-side structure, which is central to the research question. As a result, Model 4a is not used

for substantive interpretation.

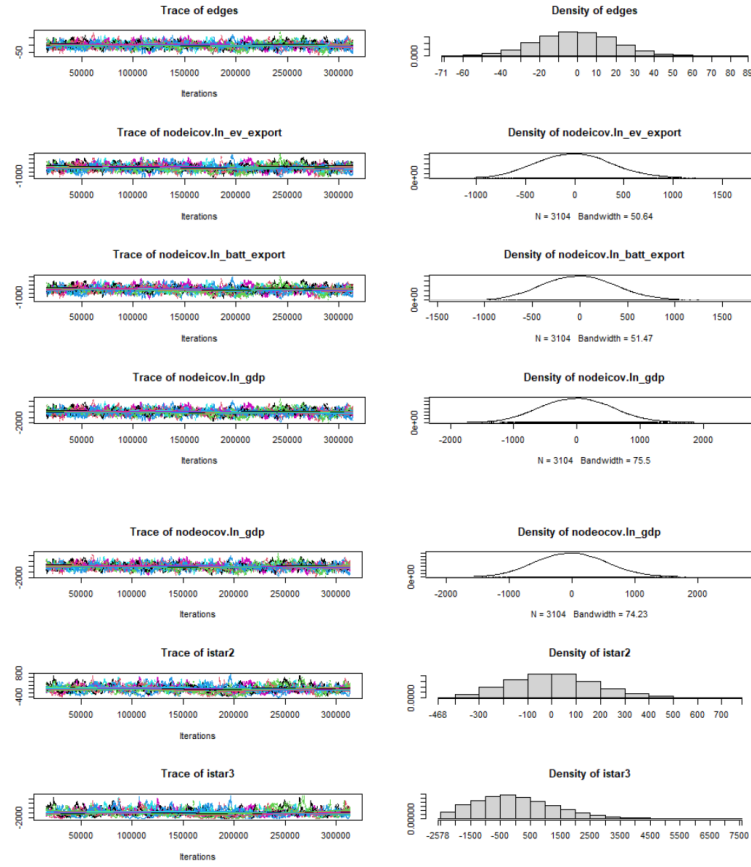


Figure 8: MCMC Diagnostics for Model 4a

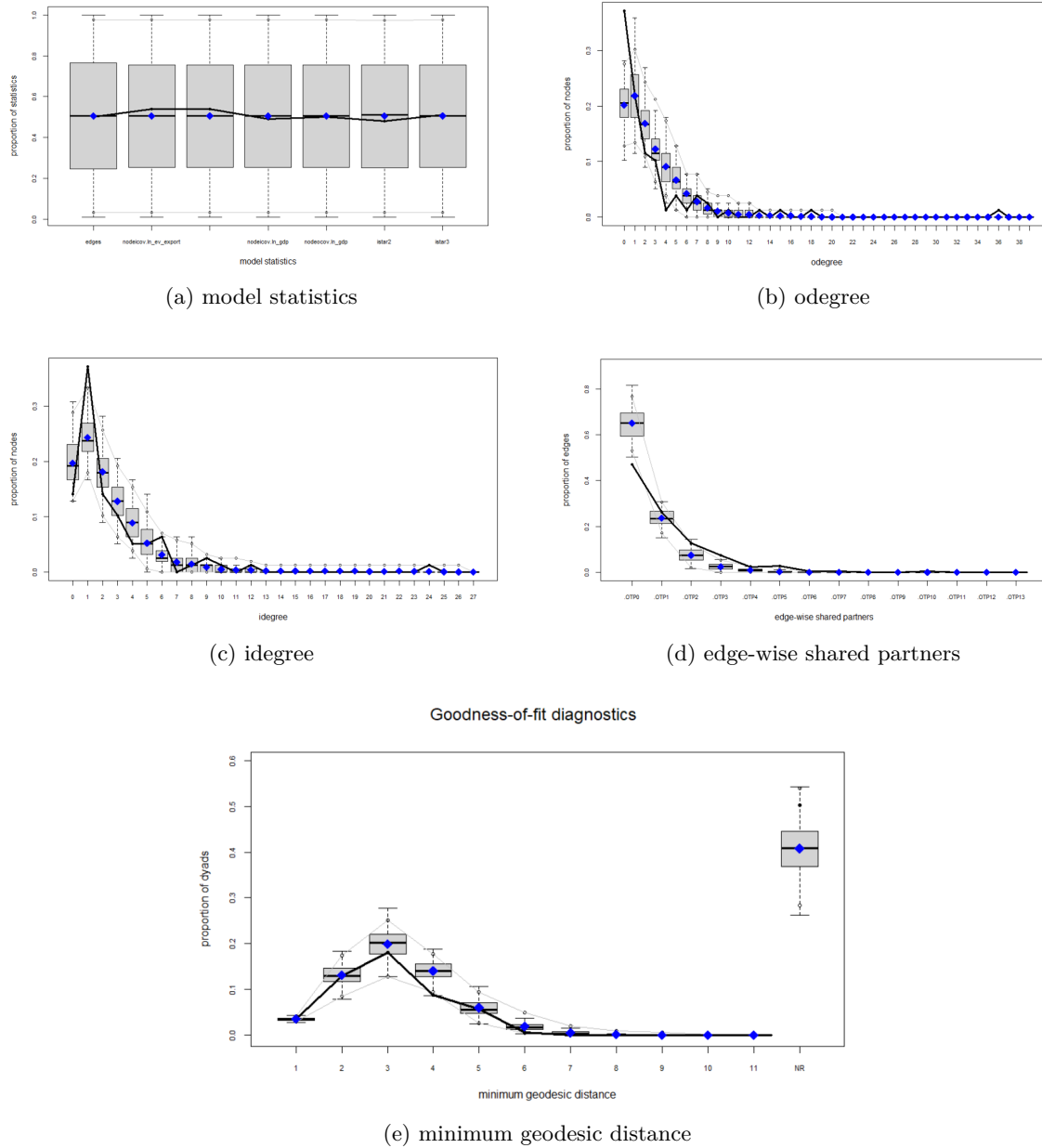


Figure 9: Goodness-of-fit (GOF) Diagnostics for Model 4a

Convergence Validation (Model 4b_extra)

To illustrate the importance of sufficient MCMC estimation effort, we estimated an auxiliary model (Model 4b_extra) that is structurally identical to Model 4b but estimated with a severely restricted number of MCMLE iterations.

In contrast to the final Model 4b, the trace plots for Model 4b_extra show poor mixing and visible trends for several parameters, most notably for the edges and ostar terms. In addition, the

corresponding MCMC sample statistics are not centered around zero, indicating that the simulated networks do not reproduce the observed statistics.

These diagnostics indicate that Model 4b_extra did not reach a stable estimation region and illustrate that sufficient iteration depth is required for reliable estimation in ERGMs with higher-order dependence terms.

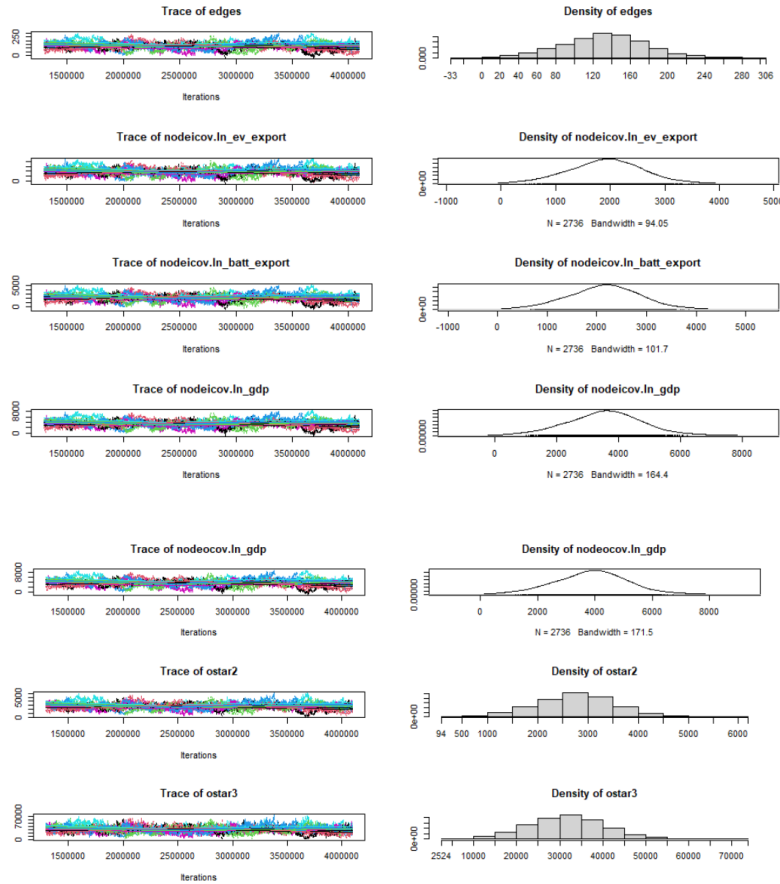


Figure 10: MCMC Diagnostics for Model 4b_extra

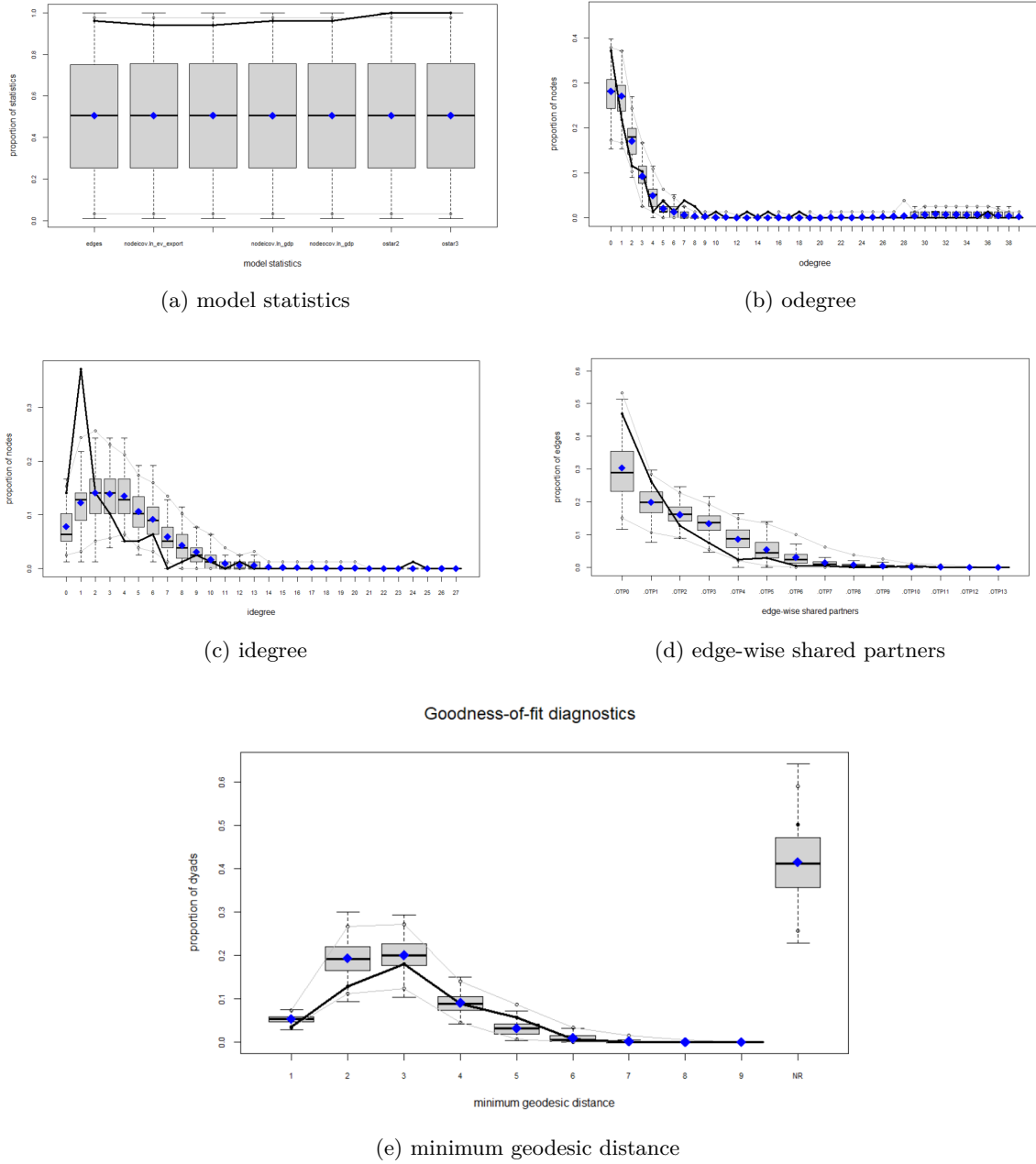


Figure 11: Goodness-of-fit (GOF) Diagnostics for Model 4b_extra

IV.4 D.4: Structural Confounding and Parameter Stability

This appendix examines how the inclusion of structural terms affects the estimated coefficients of node-level covariates. In particular, we compare Model 4a and Model 4b to assess whether changes in significance and magnitude are driven by differences in network structure rather than substantive effects.

D.4.1 Confounding in Hypothesis Variables (Model 4a vs 4b)

Model 4a and Model 4b differ in their structural specification. Model 4a includes in-star (istar) terms to model importer-side concentration, whereas Model 4b includes out-star (ostar) terms to model exporter-side concentration.

In Model 4b, the coefficients for EV exports and Battery exports are small but statistically significant, indicating that countries exporting these products are more likely to form outgoing trade ties. In Model 4a, however, both coefficients decrease in magnitude and lose statistical significance. Specifically, the EV export coefficient decreases from approximately 0.06 in Model 4b to 0.02 in Model 4a, while the Battery export coefficient decreases from approximately 0.06 to 0.03.

This change suggests that the istar terms in Model 4a absorb variation related to in-degree concentration, which overlaps with the variation explained by exporter-related covariates. As a result, the effects of EV and Battery exports become harder to identify in Model 4a.

Because Model 4a does not adequately reproduce the observed out-degree distribution (Appendix D.3), the loss of significance in this model is not interpreted as evidence against the hypotheses. Instead, Model 4b is used for hypothesis interpretation, as it better reflects export-side network structure.

D.4.1 Confounding in Control Variables (GDP)

A similar pattern is observed for GDP. In Model 3, which includes only node-level covariates, the coefficient for exporting GDP (nodecov.ln_gdp) is relatively large (approximately 0.52). In Model 4b, after including the ostar structural terms, this coefficient decreases substantially to approximately 0.08, although it remains statistically significant.

This decrease indicates that part of the effect attributed to GDP in simpler models overlaps with exporter-side network structure. Once export concentration is explicitly modeled, GDP explains less of the remaining variation in tie formation.

Overall, the comparison between Model 4a and Model 4b shows that estimated coefficients are sensitive to the inclusion of structural network terms. The results highlight that changes in significance and magnitude should be interpreted cautiously and in conjunction with model fit and GOF diagnostics. For this reason, Model 4b is retained as the preferred specification for substantive interpretation.

IV.5 D.5 Overview ERGM Models

```
model_1 <- ergm::ergm(nw_binary ~ edges,  
  control = ergm::control.ergm(  
    MCMC.burnin = 7000,  
    MCMC.samplesize = 20000,  
    seed = 123,  
    MCMLE.maxit = 40,  
    parallel = num_cores,  
    parallel.type = "PSOCK"  
  ))
```

(a) Model 1

```
model_2 <- ergm::ergm(nw_binary ~ edges +  
  nodeicov("ln_ev_export") +  
  nodeicov("ln_batt_export"),  
  control = ergm::control.ergm(  
    MCMC.burnin = 7000,  
    MCMC.samplesize = 20000,  
    seed = 1234,  
    MCMLE.maxit = 40,  
    parallel = num_cores,  
    parallel.type = "PSOCK"  
  ))
```

(b) Model 2

```
model_3 <- ergm::ergm(nw_binary ~ edges +  
  nodeicov("ln_ev_export") +  
  nodeicov("ln_batt_export") +  
  nodeicov("ln_gdp") +  
  nodecov("ln_gdp"),  
  control = ergm::control.ergm(  
    MCMC.burnin = 7000,  
    MCMC.samplesize = 20000,  
    seed = 12346,  
    MCMLE.maxit = 40,  
    parallel = num_cores,  
    parallel.type = "PSOCK"  
  ))
```

(c) Model 3

Figure 12: ERGM model specifications (Models 1–3).

```

model_4a <- ergm::ergm(nw_binary ~ edges +
  nodeicov("ln_ev_export") +
  nodeicov("ln_batt_export") +
  nodeicov("ln_gdp") +
  nodeocov("ln_gdp") +
  istar(2) +
  istar(3),
  control = ergm::control.ergm(
    MCMC.burnin = 7000,
    MCMC.samplesize = 20000,
    seed = 12349,
    MCMLE.maxit = 40,
    parallel = num_cores,
    MCMLE.steplength = 0.2,
    parallel.type = "PSOCK"
  ))

```

(a) Model 4a

```

model_4b <- ergm::ergm(nw_binary ~ edges +
  nodeicov("ln_ev_export") +
  nodeicov("ln_batt_export") +
  nodeicov("ln_gdp") +
  nodeocov("ln_gdp") +
  ostar(2) +
  ostar(3),
  control = ergm::control.ergm(
    MCMC.burnin = 7000,
    MCMC.samplesize = 20000,
    seed = 12345,
    MCMLE.maxit = 40,
    parallel = num_cores,
    MCMLE.steplength = 0.2,
    parallel.type = "PSOCK"
  ))

```

(b) Model 4b

```

model_4b_extra <- ergm::ergm(nw_binary ~ edges +
  nodeicov("ln_ev_export") +
  nodeicov("ln_batt_export") +
  nodeicov("ln_gdp") +
  nodeocov("ln_gdp") +
  ostar(2) +
  ostar(3),
  control = ergm::control.ergm(
    MCMC.burnin = 7000,
    MCMC.samplesize = 20000,
    seed = 1234578,
    MCMLE.maxit = 2,
    parallel = num_cores,
    MCMLE.steplength = 0.2,
    parallel.type = "PSOCK"
  ))

```

(c) Model 4b_extra

Figure 13: ERGM model specifications (Models 4a–4b_extra).

IV.6 D.6 Model Control Settings

To ensure reproducibility and robustness of the MCMC estimation, specific control parameters were enforced across the final models (`ergm::control.ergm`). The settings were chosen to maximize the likelihood of convergence given the complex dependency structures included in the analysis.

The specific control parameters used were:

- MCMC Burn-in: 7,000 iterations (to ensure the chain uncouples from the starting network).
- MCMC Sample Size: 20,000 (to provide a sufficient number of network statistics for estimation).
- Maximum Iterations (`MCMLE.maxit`): 40 (to allow sufficient steps for the optimization algorithm to find the maximum likelihood estimates).
- Step Length (`MCMLE.steplength`): 0.2 (for the final models 4a and 4b a conservative step length was chosen to prevent the optimization from overshooting for the sensitive star terms).
- Parallel Processing: Utilizing PSOCK clusters to distribute computation.
- Seed: Fixed seeds (e.g., 12345) were used for each model run to ensure exact reproducibility.

IX. TECHNOLOGY STATEMENT

During the preparation of this work, We used ChatGPT and Gemini in order to brainstorm research directions, refine the scope, organize the report structure, clarify academic terminology, identify relevant literature themes, and receive feedback on writing clarity and coherence. The following parts of the assignment were affected/generated by AI tool usage: Abstract, Introduction, Research Rationale, Dataset, Results, and Conclusion. Furthermore, the code was refined by using AI. After using this tool/service, Lars Kerkhof, Ralf Geerts, Riccardo Festa, Vera Beerepoot, and Yvette Garretsen evaluated the validity of the tool's outputs, including the sources that generative AI tools have used, and edited the content as needed. As a consequence, Lars Kerkhof, Ralf Geerts, Riccardo Festa, Vera Beerepoot, and Yvette Garretsen take full responsibility for the content of their work.